Individual assistance during Large Disasters - based on data from OpenFEMA

Introduction

This report focuses on what insights can benefit insurance companies from data gathered by the Federal Emergency Management Agency (FEMA) during Large Disasters.
Insurance corporations sell various kinds of policies, helping homeowners and renters alike, to protect their investment. While there is no law mandating home insurance, it can provide a home owner or a renter peace of mind protecting their investment (Home insurance isn't required. But it makes sense. (2021). Texas Department of Insurance).
Disasters can be man-made or by an act of nature. FEMA maintains a list of disasters along with summaries. Once a disaster is declared by the President of the United States as a Major Disaster (FEMA. (2022)), FEMA also maintains a list of Individual registrants and the damage assessments incurred by the registrant.
Insights from Individual Registrants, Declarations and Summaries could potentially help Insurance companies to enhance their Risk Assessment, better management of claims and market expansion.

Data

Data used for this analysis is from FEMA (under Reports & Data/OpenFEMA/Data Sets) and belong to 3 datasets:

1. Individual Assistance Housing Registrants (Large Disasters): Data here is collected by FEMA of all individual registrants during Large Disasters (e.g., Hurricanes, Severe Winter Storms). Data such as location - damagedCity, damagedZipCode, insurance status - homeOwnersInsurance, floodInsurance, Damage Assessment - foundation damage, flood damage, real property damage assessed, destroyed, special needs are all captured for an individual. Each individual is then associated with a **DisasterNumber**, which can then be mapped with the next 2 tables, which can help in generating meaningful insights.

Total records	6367334
Parameters	42
CSV file size	1 GB
Null Values handling	highWaterLocation and rental assistance related information contain null values for more than 99%, hence not used for analysis

2. FEMA Disaster Declarations: Data here is also collected by FEMA and it focuses more on the Date values of when the disaster occurred - incidentBeginDate, incidentEndDate, disasterDeclarationDate and various Assistance Programs declared for each disaster. This table also contains DisasterNumber which is helpful in linking with other tables used for this analysis.

Total records	5061
Parameters	26
CSV file size	2.4 MB
Null Values handling	shapeFileURL, kmzFileURL contain nulls for more than 75% of data. These are not being used for analysis.

3. FEMA Disaster Summaries: Data here, is also collected by FEMA, and contains counts of Approved for assistance applications and the amounts obligated, approved for each disaster. Financial information is derived from NEMIS and not FEMA's official financial system

Total records	3817
Parameters	14
CSV file size	619.3 KB
Null Values handling	Some of the total approved amount columns contain more than 85% of null values, but when filtered for large disasters these amounts are captured. Thus querying parameters seem to be helping filter this out.

For the data dictionary, please visit the Appendix section - Appendix/Data Dictionary
For data-procuring and cleaning process visit the Appendix section - Appendix/Data Pipelining Process.

Analysis and results

Data on Individual Registrants, Disaster Summary and Disaster Declarations were obtained from OpenFEMA datasets in their comma separated values format. Majority of the data gathering and processing was performed using Google's Cloud Platform. Sequence of steps are as follows:

- For Data extraction, Google's compute engine was used to spin up high capacity virtual machines
- terminal commands were executed to copy the CSV files into the VM's storage
- CSV files were then copied to Storage Buckets within Google Cloud Platform
- Data was then imported into BigQuery Dataset and tables

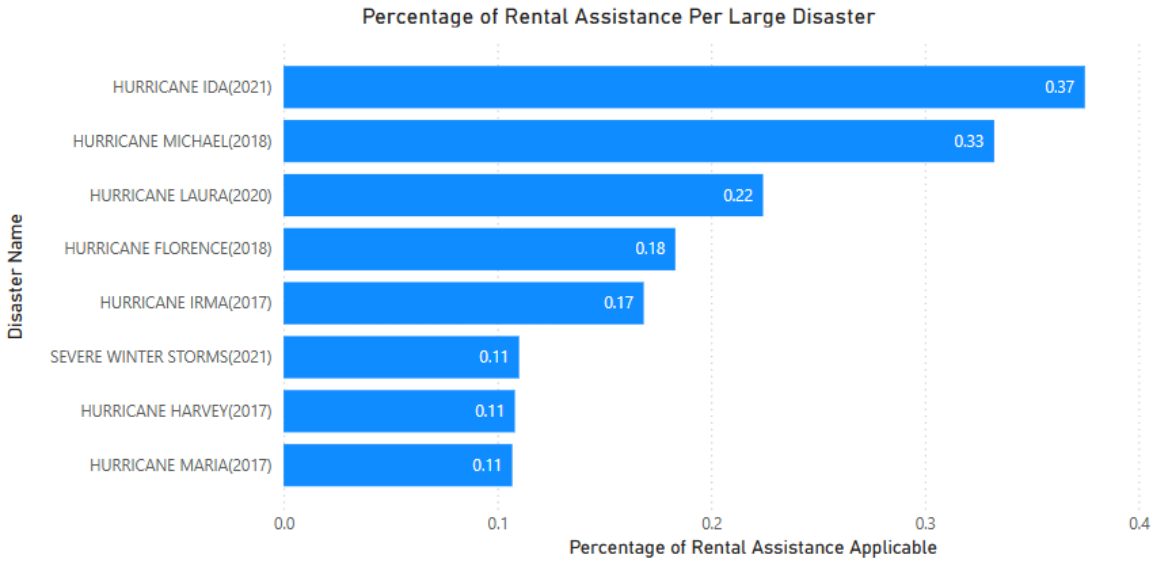
- Insights were generated using SQL queries from the data in these tables

Data from Individual Registrants had information on the damages incurred and their amounts. It also recorded the eligibility of individual registrants toward various assistance programs. In conjecture with Disaster Summary and Declaration data, here are some insights that were considered for interpretation:

- What is the percentage of renters who have applied and compensated for during Large disasters?
  Result: 37.48% of renters during Hurricane Ida in 2021 received rental assistance, which is recorded as the highest amount of renters being declared eligible.
- Where is the concentration of Zip Codes which has significant losses to personal property during these Large Disasters?
  Result: Highest concentration of Zip Codes with Significant Losses (more than 10 times the Average of Verified Personal Property losses) was in the State of Louisiana during Hurricane Laura.
- How do Large Disasters affect the structural integrity in each residence type?
  Result: The trends of roof damage and foundation damage amounts seem to be correlating with each other during Large Disasters, for each structure type.
- Are there any outliers with extremely high housing assistance beneficiaries?
  Result: Extremely high assistance amounts were provided during Hurricanes Katrina in Louisiana and Harvey in Texas (more than 95 percentile)
- Till when are the registrants allowed to seek assistance from FEMA for a disaster?
  Result: Various disaster types have their average time to close out period since they were declared as a Disaster. In the instances of Tsunami and Earthquakes have accepted applicants for assistance as late as 10 years from the incident date.

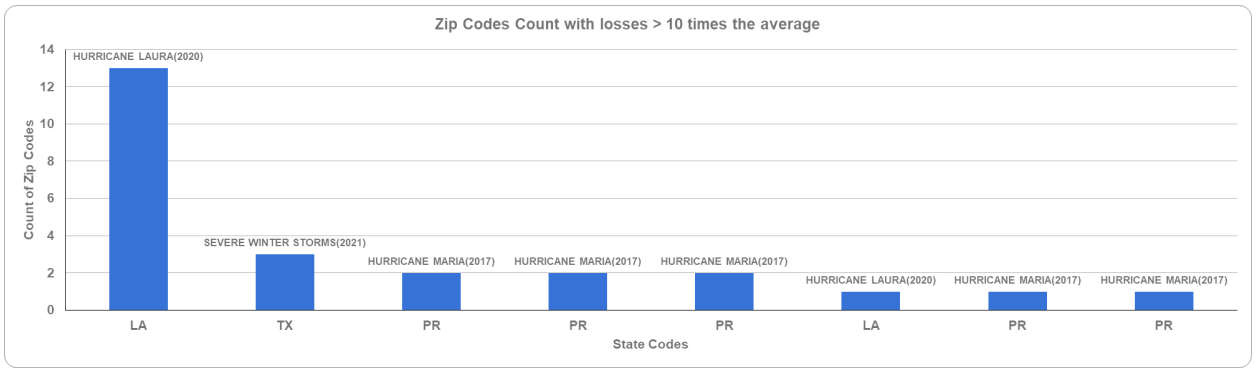
Discussion(interpretation)

Rental Assistance for Registrants:



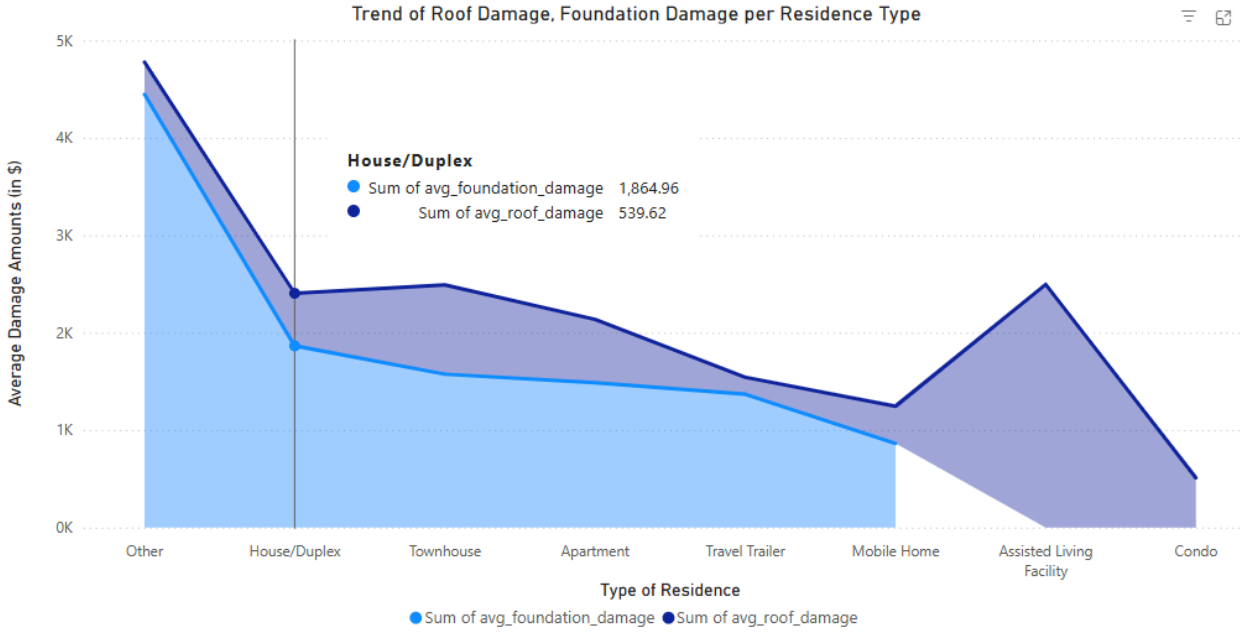
While owners are covered under various assistance programs including SBA assistance, renters have lesser options. Even the repair amounts are significantly lesser than the owners for renters. Highest that was ever compensated being 37.48% of the renters who applied for assistance during Hurricane Ida in 2021.
Insurance Companies could potentially view this as an opportunity to reach out to residents of the regions. This could also be interpreted as to avoid such a renter demography to lessen losses meaning when assistance is not provided by the Government may lead the customers to reach out to insurance companies for payout.

Highest concentration of Zip Codes with significant losses to personal property (PPFVL):



PPFVL is FEMA verified losses to Personal Property. These losses are more than 10 times the average losses of all registrants, thus are very significant. Insurance companies may need to carefully strategise their premia in these Zip Codes to minimise losses. Also, to be careful in designing policies since historically the damages incurred are significant.

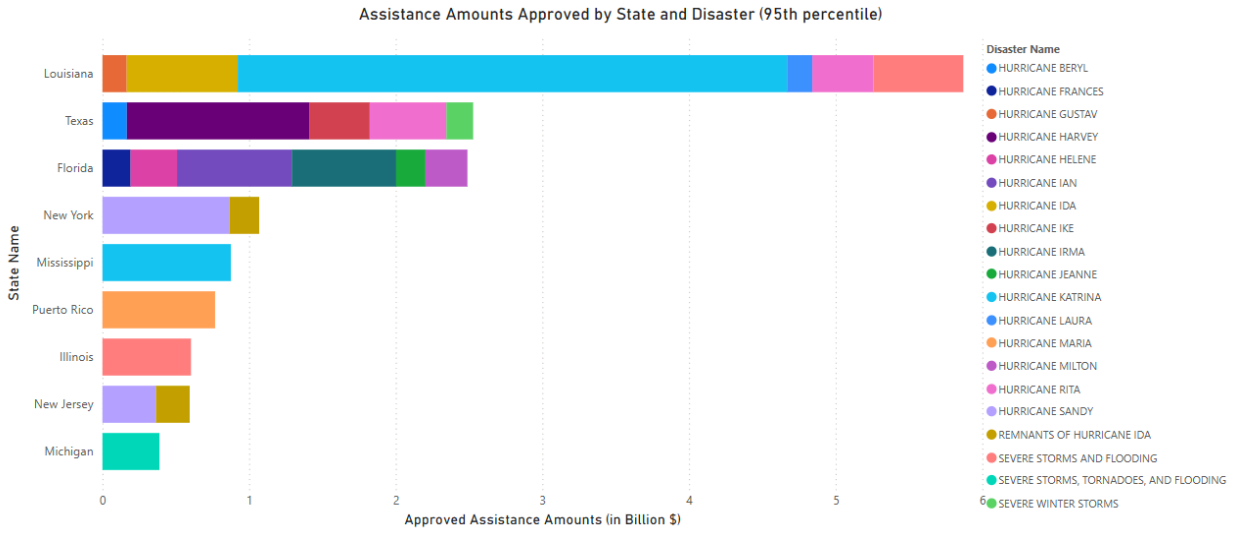
Trend between damages across structure types:



This focuses more on structural integrity in various residence or structure types. A trend plot of roof damage amounts and foundation damages reveals that they seem to follow each

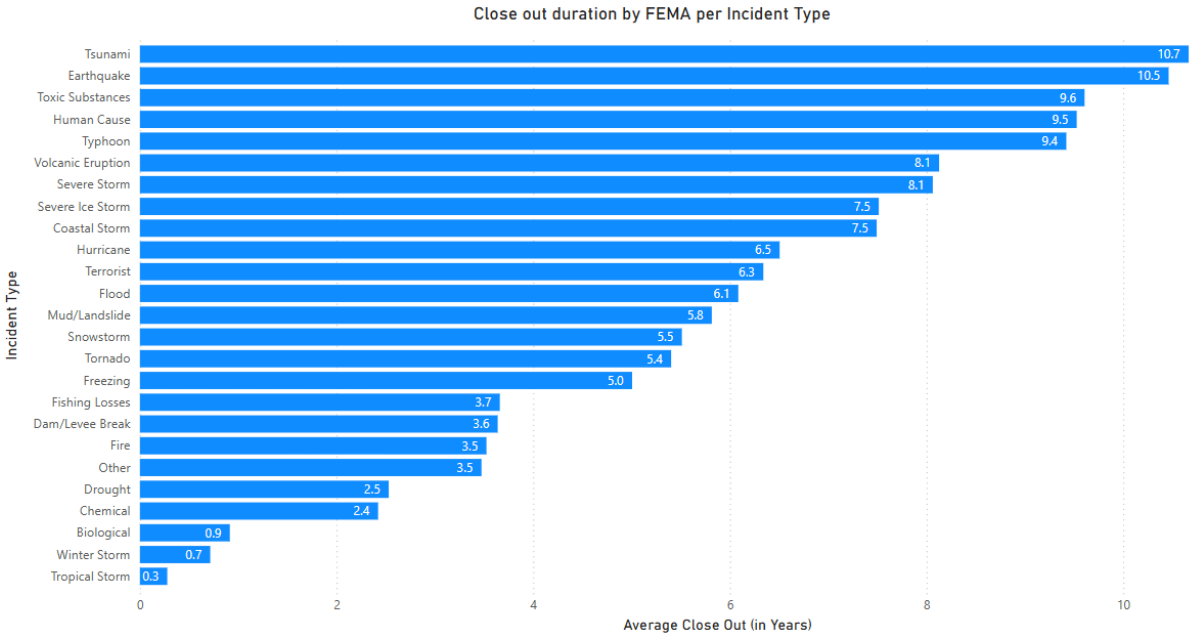
other closely, during a Large disaster. Insurance companies may not exactly benefit from this information since it does not clearly distinguish between the two. There is one key takeaway from this is that during a Large Disaster there is a correlation. So, policies can be framed with suitable verbiage to compensate for damages during Large disasters based on this inference.

Outliers in Housing Assistance recipients:



Some regions have been fortunate enough to receive generous assistance from the Government. These are the states above 95th percentile in the amount of assistance received. Clauses and the premia could be suitably drafted for policies being issued in these areas since the government assistance is high.

Average Closeout times for various Incident types:



FEMA closeout dates are the cut off dates after which applications are no longer accepted for assistance. Some of them are well beyond 8 years since the incident occurred. FEMA assistance if applicable can be used in lieu of insurance payout and vice versa. Careful scrutiny of declarations can help registrants claim FEMA assistance rather than from Insurance companies. Registrants can be checked if they qualify for assistance from FEMA before the payout from Insurance companies.

Conclusion

Insurance companies could benefit by careful understanding of regulations from FEMA for devising their Catastrophe modeling to premia calculation. Some of these are as follows:

- Hurricanes seem to be leading Large disaster and seem to frequently affect registrants of states Louisiana, Texas and Puerto Rico
- During a large disaster the damage level is destroyed for most properties of type House/Duplex and damages seem similarly affecting roof and foundations
- There are notable instances where registrants are compensated extensively, thus check for Assistance provided by the Government in regions like Louisiana, Texas and Florida before crafting policies for the residents

While this analysis was done aiming to help Insurance companies while making policy guidelines based on region and type of disaster, this can also be used by the residents of the areas where disaster has affected previously. This could provide an insight on how the assistance programs have been provided in the past for these areas.

References

Home insurance isn't required. But it makes sense. (2021). Texas Department of Insurance. https://www.tdi.texas.gov/blog/home-insurance-isnt-required-but-makes-sense.html

FEMA. (2022). How a disaster gets declared. Www.fema.gov. https://www.fema.gov/disaster/how-declared

Appendix

Link to GCP project: udvramesh
Link to Presentation: VRAMESH_Insights from FEMA Data

Data Pipelining Process:
Tools used are

1. Data extraction - Google's Compute engine
    1.1. Create a VM on GCP using required compute performance settings
    1.2. Launch SSH terminal for the next set of commands
    1.3. Loading data from FEMA (OpenFEMA links)
        1.3.1. Individual Registrants dataset : wget https://www.fema.gov/about/reports-and-data/openfema/IndividualAssistanceHousingRegistrantsLargeDisasters.csv

1.3.2. Disaster declaration dataset: wget https://www.fema.gov/api/open/v1/FemaWebDisasterDeclarations.csv

1.3.3. Disaster Summary dataset: wget https://www.fema.gov/api/open/v1/FemaWebDisasterSummaries.csv

2. Storage - Cloud Storage

Type the following commands on the SSH terminal of the VM

Create a Bucket with the necessary storage parameters. Bucket created here is named term_project_open_fema

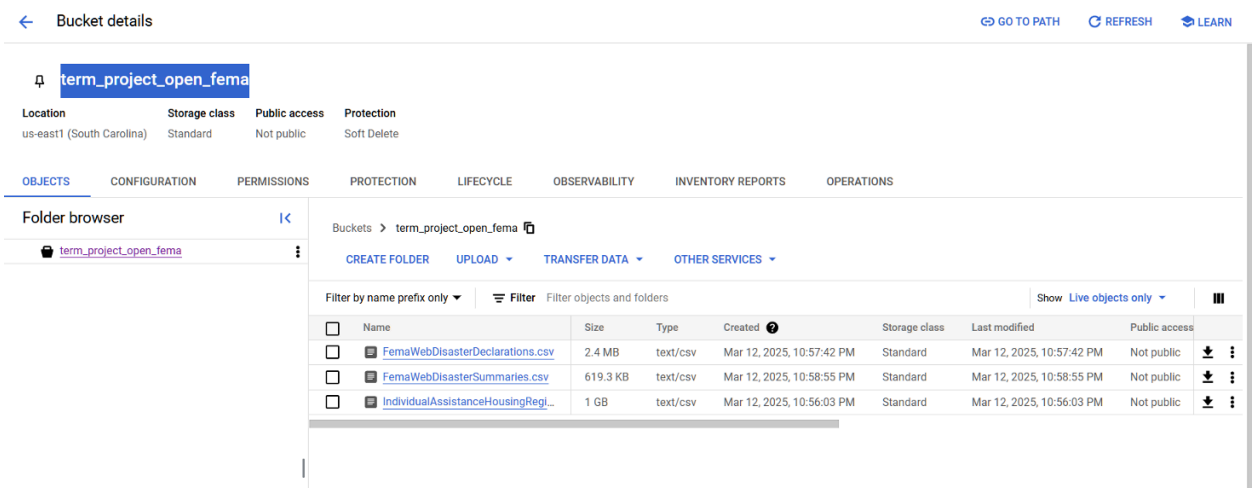
2.1. Loading Individual registrant dataset to GCP: gsutil cp IndividualAssistanceHousingRegistrantsLargeDisasters.csv gs://term_project_open_fema/IndividualAssistanceHousingRegistrantsLargeDisasters.csv

2.2. Loading Disaster Declaration dataset to GCP: gsutil cp FemaWebDisasterDeclarations.csv gs://term_project_open_fema/FemaWebDisasterDeclarations.csv

2.3. Loading Disaster Summary dataset to GCP: gsutil cp FemaWebDisasterSummaries.csv gs://term_project_open_fema/FemaWebDisasterSummaries.csv

3. Data Transformation - Google's BigQuery

By the end of step-2, all the CSV files should be loaded in the created bucket and looking something like this



Once the data is in the Storage Bucket, they need to be transferred into a Dataset and then further into their individual tables.

Dataset created under project is : Dataset_OpenFEMA

Create individual tables from each of the loaded csv files in the Storage Bucket.

Create table

Source

Create table from
Google Cloud Storage

Select file from GCS bucket or use a URI pattern *

File format
Avro

Source Data Partitioning

Destination

Project *
udvramesh

Dataset *
Dataset_OpenFEMA

Table *

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors,

Table type
Native table

Schema

Source file defines the schema.

Partitioning settings

No partitioning
Partition by ingestion time
Partition by field
To partition by field, you must first add fields to the schema above.

Cluster settings

Clustering order

Clustering order determines the sort order of the data. Clustering can be used on both pa

Tags

Tags help you manage and enforce policies on your resources. Tags consist of a unique ta
and a set of tag values. Learn more

Select scope

Advanced options

Create table Cancel

Choose a file

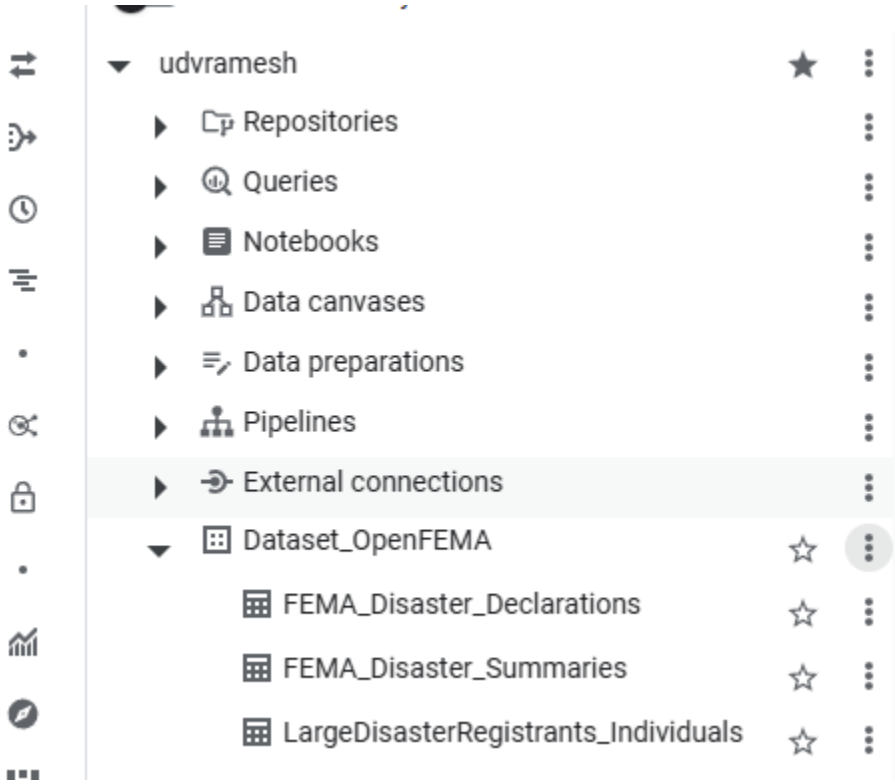
term_project_open_fema

FemaWebDisasterDeclarations.csv

FemaWebDisasterSummaries.csv

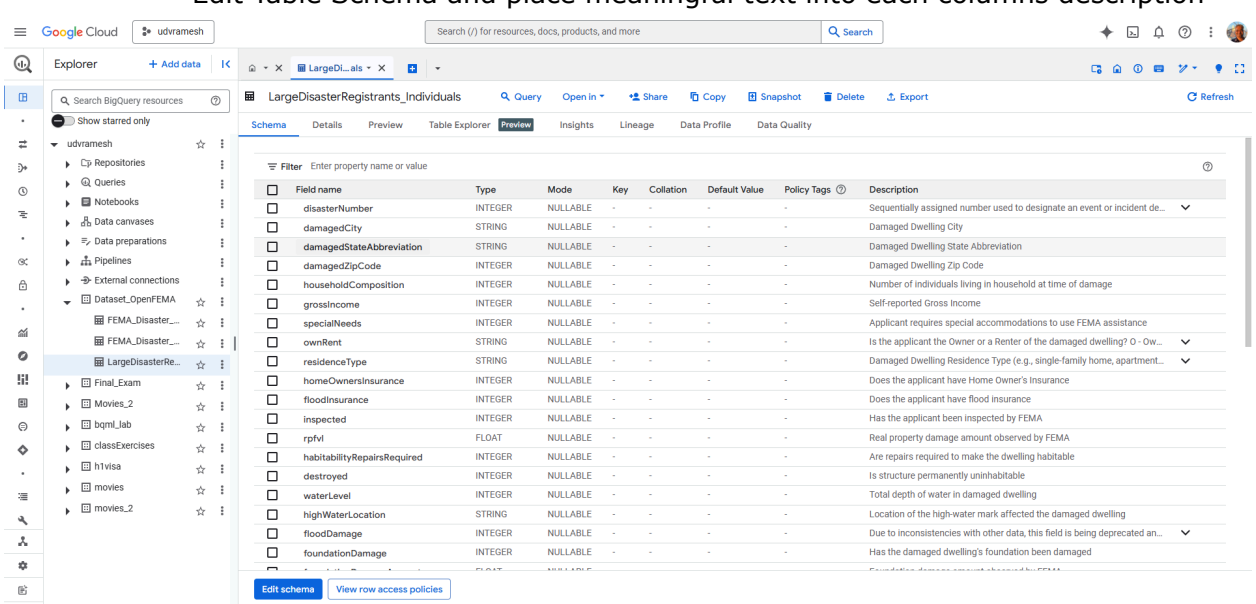
IndividualAssistanceHousingRegistrantsLargeDisasters.csv

Once all tables are created they should look something like this:

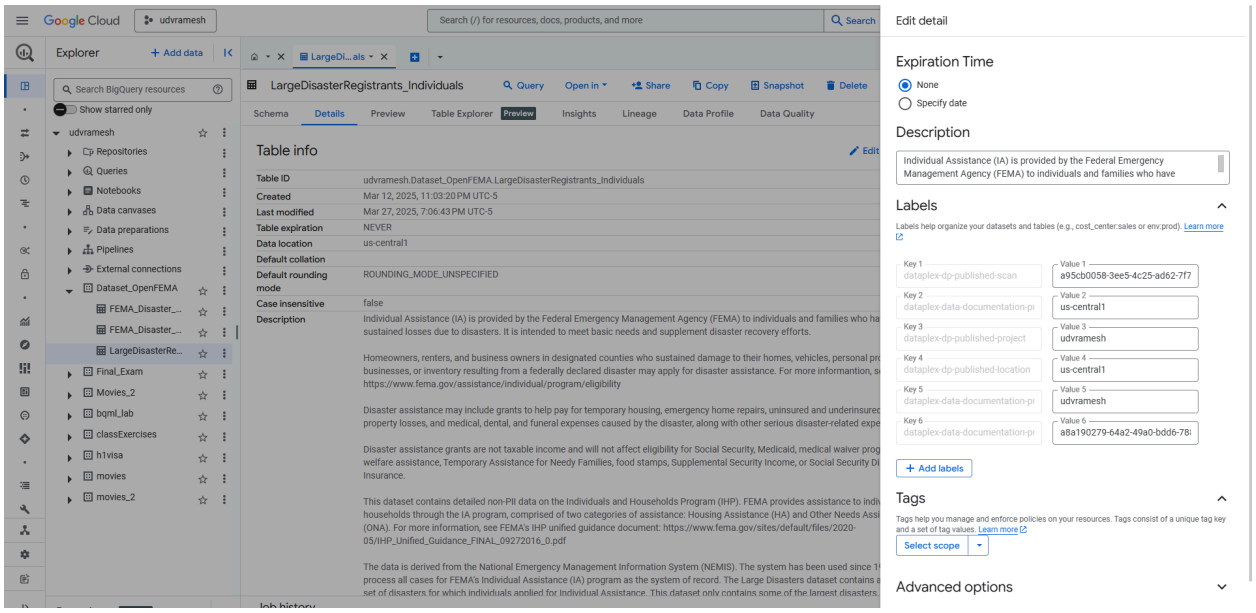


4. [Optional] For each of these tables, create a Data Profile, which can then be used to generate Insights. This is a feature in BigQuery which uses the description entered for the table and each of their columns and generate insights using Google's Gemini AI

4.1. Adding column and table descriptions:
Edit Table Schema and place meaningful text into each columns description

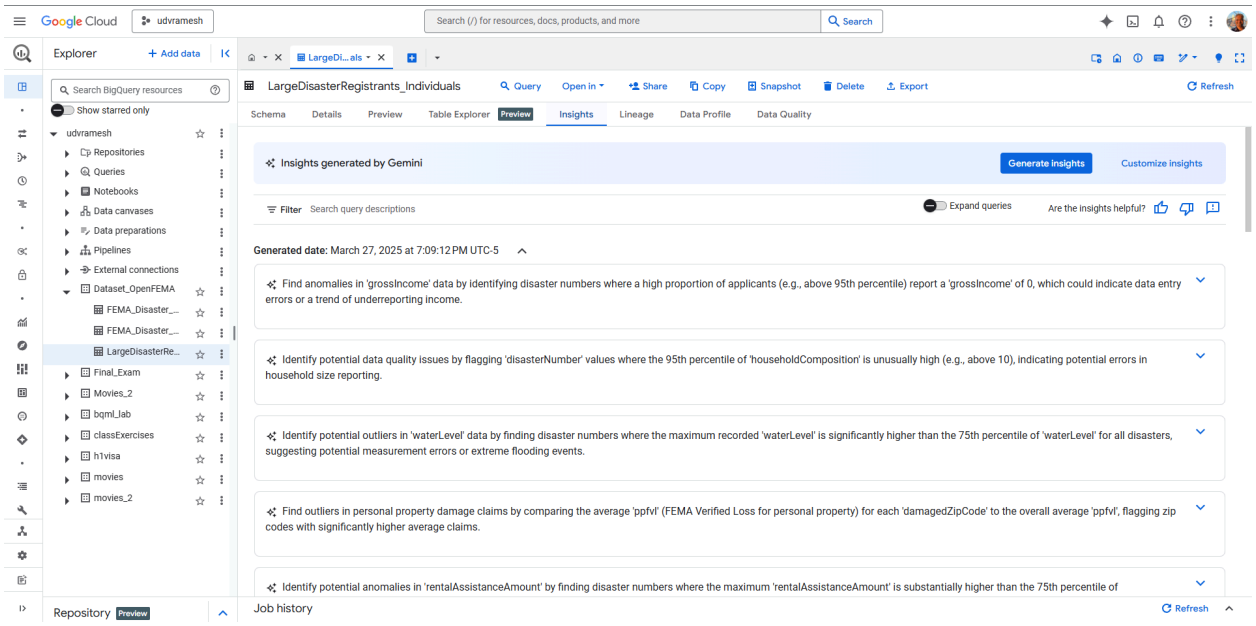


4.2. Add table **Description** under Details section



4.3. Generated Insights:

Under the section Insights, click to generate insights. This might take a while. Once the process is complete this might look something like this:



5. Visualization - [Google Sheets]() and [Looker Studio]():

These contain the links to the Google Sheets to where I extracted the queried data to generate visualizations

Queries:

```
-- Calculate the percentage of homeowners versus renters who received repair
assistance for each disaster
SELECT
  CONCAT(disasterName,'(',EXTRACT(YEAR FROM DD.declarationDate),')') AS disasterName,
  -- ownRent,
  -- (COUNTIF(repairAssistanceEligible = 1) * 100.0 / COUNT(*)) AS
percentageRepairAssistance,
  -- (COUNTIF(replacementAssistanceEligible = 1) * 100.0 / COUNT(*)) AS
percentageReplacementAssistance,
  (COUNTIF(rentalAssistanceEligible = 1) / COUNT(*)) AS
percentageRentalAssistanceEligible,
  SUM(rentalAssistanceAmount) / 1000000 AS totalRentalAssistantAmount,
  -- EXTRACT(YEAR FROM DD.incidentEndDate)
FROM
  `Dataset_OpenFEMA.LargeDisasterRegistrants_Individuals` IR
  INNER JOIN `udvramesh.Dataset_OpenFEMA.FEMA_Disaster_Declarations` DD ON
IR.disasterNumber = DD.disasterNumber
WHERE
  ownRent IN ('Renter')
GROUP BY
  disasterName
  -- , incidentEndDate
ORDER BY 2 DESC;

-- Find outliers in personal property damage claims by comparing the average 'ppfvl'
(FEMA Verified Loss for personal property) for each 'damagedZipCode' to the overall
average 'ppfvl', flagging zip codes with significantly higher average claims.
SELECT
  -- damagedZipCode,
  -- damagedCity,
  DISTINCT(damagedStateAbbreviation),
  COUNT(damagedZipCode) count_of_zipcodes,
  CONCAT(disasterName,'(',EXTRACT(YEAR FROM DD.declarationDate),')') AS disasterName,
  AVG(ppfvl) AS avg_ppfvl_by_zip_code
FROM
  `Dataset_OpenFEMA.LargeDisasterRegistrants_Individuals` IR
  INNER JOIN `udvramesh.Dataset_OpenFEMA.FEMA_Disaster_Declarations` DD ON
IR.disasterNumber = DD.disasterNumber
GROUP BY
  disasterName,
  damagedZipCode,
  -- damagedCity,
  damagedStateAbbreviation
HAVING
  avg_ppfvl_by_zip_code > (
  SELECT
    AVG(ppfvl)
  FROM
    `Dataset_OpenFEMA.LargeDisasterRegistrants_Individuals` ) * 10
ORDER BY 2 DESC;

-- What is the average roof damage amount vs foundation for each residence type?
SELECT
```

```sql
  COUNT(damagedZipCode) AS count_of_damagedZipcodes,
  -- waterLevel,
  residenceType,
  -- COUNT(roofDamage) count_roofDamage,
  -- COUNT(foundationDamage) count_foundationDamage,
  COUNT(destroyed) count_destroyed,
  -- damagedStateAbbreviation,
  AVG(roofDamageAmount) avg_roof_damage,
  AVG(foundationDamageAmount) avg_foundation_damage,
  -- AVG(roofDamageAmount) * 100 / COUNT(damagedZipCode) percent_roof_damage,
  -- AVG(foundationDamageAmount) * 100 / COUNT(damagedZipCode)
percent_foundation_damage
FROM
  `Dataset_OpenFEMA.LargeDisasterRegistrants_Individuals`
WHERE
  waterLevel > 0
  AND (roofDamageAmount > 0
  OR foundationDamageAmount > 0)
GROUP BY
  -- waterLevel,
  residenceType
  -- damagedZipCode
  -- damagedStateAbbreviation
-- HAVING
--   avg_roof_damage > 0
--   AND avg_foundation_damage > 0
ORDER BY
  count_destroyed DESC;

SELECT
  incidentType,
  AVG(DATE_DIFF(closeoutDate, incidentEndDate, DAY)/365) AS avg_time_to_close
FROM
  `Dataset_OpenFEMA.FEMA_Disaster_Declarations` DD
WHERE
  incidentEndDate IS NOT NULL
  AND closeoutDate IS NOT NULL
AND DATE_DIFF(closeoutDate, incidentEndDate, DAY) > 0
GROUP BY
  incidentType
ORDER BY
  avg_time_to_close DESC;

-- This query identifies disasters where the 'totalAmountHaApproved' exceeds the 95th
percentile, indicating extremely high housing assistance amounts.
SELECT
  -- DS.disasterNumber,
  disasterName,
  stateName,
  SUM(totalAmountHaApproved)/1000000000 HaApproved_inMil
  -- APPROX_QUANTILES(totalAmountHaApproved, 100)
FROM
  `Dataset_OpenFEMA.FEMA_Disaster_Summaries` DS
```

```
  INNER JOIN `udvramesh.Dataset_OpenFEMA.FEMA_Disaster_Declarations` DD ON
DS.disasterNumber = DD.disasterNumber
WHERE
  totalAmountHaApproved > (
  SELECT
    APPROX_QUANTILES(totalAmountHaApproved, 100)[
  OFFSET
    (95)]
  FROM
    `Dataset_OpenFEMA.FEMA_Disaster_Summaries`)
GROUP BY
  stateName,
  disasterName
ORDER BY
    3 DESC;
```

Data dictionary:
[Individual Registrants]

Name	Title	Type	Description	Is Searchable
disasterNumber	Disaster Number	smallint	Sequentially assigned number used to designate an event or incident declared as a disaster. For more information on the disaster process, https://www.fema.gov/disasters/how-declared	yes
damagedCity	Damaged City	text	Damaged Dwelling City	yes
damagedStateAbbreviation	Damaged State Abbreviation	text	Damaged Dwelling State Abbreviation	yes
damagedZipCode	Damaged Zip Code	text	Damaged Dwelling Zip Code	yes

householdComposition	Household Composition	text	Number of individuals living in household at time of damage	yes
grossIncome	Gross Income	text	Self-reported Gross Income	yes
specialNeeds	Special Needs Indicator	boolean	Applicant requires special accommodations to use FEMA assistance	no
ownRent	Owner or Renter	text	Is the applicant the Owner or a Renter of the damaged dwelling? O - Owner; R - Renter	yes
residenceType	Residence Type	text	Damaged Dwelling Residence Type (e.g., single-family home, apartment). A - Apartment; B - Boat; C - Condo; H - House/Duplex; M - Mobile Home; N - Non-Traditional; O - Other; TH - Townhouse; T - Travel Trailer; ALF - Assisted Living Facility; MIL - Military Housing; CD - College Dorm; CF - Correctional Facility	yes
homeOwnersInsurance	Home Owners Insurance	boolean	Does the applicant have Home Owner's Insurance	no
floodInsurance	Flood Insurance Indicator	boolean	Does the applicant have flood insurance	no
inspected	Inspected Indicator	boolean	Has the applicant been inspected by FEMA	yes
rpfvl	RPFVL	decimal	Real property damage amount observed by FEMA	yes

habitabilityRepairsRequired	Habitability Repairs Required	boolean	Are repairs required to make the dwelling habitable	yes
destroyed	Destroyed	boolean	Is structure permanently uninhabitable	no
waterLevel	Water Level	smallint	Total depth of water in damaged dwelling	yes
highWaterLocation	High Water Location	text	Location of the high-water mark affected the damaged dwelling	yes
floodDamage	Flood Damage Indicator	boolean	Due to inconsistencies with other data, this field is being deprecated and will no longer appear in this data set	no
foundationDamage	Foundation Damage Indicator	boolean	Has the damaged dwelling's foundation been damaged	no
foundationDamageAmount	Foundation Damage Amount	decimal	Foundation damage amount observed by FEMA	yes
roofDamage	Roof Damage Indicator	boolean	Has the damage dwelling's roof been damaged	no
roofDamageAmount	Roof Damage Amount	decimal	Roof damage amount observed by FEMA	yes
tsaEligible	Temporary Sheltering	boolean	Is applicant eligible for Temporary Sheltering Assistance	no

	Assistance Eligible			
tsaCheckedIn	TSA Checked In	boolean	Has applicant checked in to FEMA provided Temporary Sheltering Assistance facility	no
rentalAssistanceEligible	Rental Assistance Eligible	boolean	Is applicant eligible for FEMA rental assistance	no
rentalAssistanceAmount	Rental Assistance Amount	decimal	Amount of Rental Assistance in dollars	yes
repairAssistanceEligible	Repair Assistance Eligible	boolean	Is applicant eligible for FEMA assistance to repair the damaged dwelling	no
repairAmount	Repair Amount	decimal	Amount of Repair Assistance in dollars	yes
replacementAssistanceEligible	Replacement Assistance Eligible	boolean	Is applicant eligible for FEMA assistance to replace the damaged dwelling	no
replacementAmount	Replacement Amount	decimal	Amount of Replacement Assistance in dollars	yes
sbaEligible	Small Business Association Eligible	boolean	Is applicant eligible for a Small Business Association loan	no

renterDamageLevel	Renter Damage Level	text	Level of Damage: Moderate, Major, Destroyed	yes
rentalAssistanceEndDate	Rental Assistance End Date	date	Final Month applicant received Rental Assistance	yes
rentalResourceCity	Rental Resource City	text	Rental Resource City	yes
rentalResourceStateAbbreviation	Rental Resource State Abbreviation	text	Rental Resource State Abbreviation	yes
rentalResourceZipCode	Rental Resource Zip Code	text	Rental Resource Zip Code	yes
primaryResidence	Primary Residence	boolean	Is the applicant's damaged dwelling his/her primary residence	no
personalPropertyEligible	Personal Property Eligible	boolean	Is the applicant eligible for FEMA's Other Needs Assistance (ONA) to cover damaged personal property	no
ppfvl	Personal Property Verified Loss	decimal	FEMA Verified Loss captured during inspection of personal property	yes

censusBlockId	Census Block ID	text	Address-based 15-character code that is the concatenation of fields consisting of the 2-character state FIPS code, the 3-character county FIPS code, the 6-character census tract code, and the 4-character tabulation block code. Please see: https://www.census.gov/geographies/reference-files/time-series/geo/block-assignment-files.html CENSUS BLOCK ID MAY HAVE TO BE RE-FORMATTED IN EXCEL TO SEE THE ID CORRECTLY. CREATE A CUSTOM DATA TYPE WITH FIFTEEN HASH SYMBOLS: ###############	yes
censusYear	Census Year	smallint	Census period used to obtain Census Block ID	no
id	ID	uuid	Unique ID assigned to the record	yes

Disaster Declarations

Name	Title	Type	Description	Is Searchable
disasterNumber	Disaster Number	smallint	Sequentially assigned number used to designate an event or incident declared as a disaster	yes
declarationDate	Declaration Date	date	The date the disaster was declared	yes
disasterName	Disaster Name	text	The proper name or description of the disaster, i.e. HURRICANE MARIA, FLOODING, SEVERE STORMS	yes
incidentBeginDate	Incident Begin Date	date	The date the incident itself began	yes

incidentEndDate	Incident End Date	date	The date the incident itself ended	yes
declarationType	Declaration Type	text	Defines whether a disaster is a Major Disaster Declaration, Emergency Declaration, Fire Management Assistance Declaration, or Fire Suppression Authorization	yes
stateCode	State Code	text	Two letter code for U.S. states, districts, and territories	yes
stateName	State Name	text	The name or phrase describing the U.S. state, district, or territory	yes
incidentType	Incident Type	text	The primary or official type of incident such as a fire or flood. Secondary incident types may have been designated. See the designatedIncidentTypes field. For more information on incident types, please visit https://www.fema.gov/disasters/how-declared	yes
entryDate	Entry Date	date	The date the declared disaster was entered into NEMIS	yes
updateDate	Update Date	date	The date the disaster declaration data was last updated by NEMIS	yes
closeoutDate	Closeout Date	date	The official closeout date of a disaster, after which, one can no longer apply for assistance	yes
region	Region	smallint	FEMA region number	yes
ihProgramDeclared	IH Program Declared	boolean	Denotes whether the Individuals and Households program was declared for this disaster. For more information on	yes

			the program, please visit https://www.fema.gov/assistance/individual/program	
iaProgram Declared	IA Program Declared	boolean	Denotes whether the Individual Assistance program was declared for this disaster. For more information on the program, please visit https://www.fema.gov/assistance/individual/program	yes
paProgram Declared	PA Program Declared	boolean	Denotes whether the Public Assistance program was declared for this disaster. For more information on the program, please visit https://www.fema.gov/assistance/public/program-overview	yes
hmProgramDeclared	HM Program Declared	boolean	Denotes whether the Hazard Mitigation program was declared for this disaster. For more information on the program, please visit https://www.fema.gov/grants/mitigation	yes
designatedIncidentTypes	Declared Incident types	text	A comma-separated list of incident types designated for the disaster. The primary incident type is described in the incidentType field. Codes are: 0: Not applicable; 1: Explosion; 2: Straight-Line Winds; 3: Tidal Wave; 4: Tropical Storm; 5: Winter Storm; 8: Tropical Depression; A: Tsunami; B: Biological; C: Coastal Storm; D: Drought; E: Earthquake; F: Flood; G: Freezing; H: Hurricane; I: Terrorist; J: Typhoon; K: Dam/Levee Break; L: Chemical; M: Mud/Landslide; N: Nuclear; O: Severe Ice Storm; P: Fishing Losses; Q: Crop Losses; R: Fire; S: Snowstorm; T: Tornado; U: Civil Unrest; V: Volcanic Eruption; W: Severe Storm; X: Toxic Substances; Y: Human Cause; Z: Other	yes
declarationRequestDate	Declaration Request Date	date	Date when the declaration request was made	yes

disasterPageUrl	FEMA Disaster Page	text	A URL that points to the FEMA Disaster page. Each page provides details such as funding obligations, designated areas, and other local resources for each disaster.	yes
shapefileUrl	Shapefile URL	text	A URL that points to a geospatial shapefile for the disaster. Available from 2010 to present.	yes
kmzfileUrl	KMZ File URL	text	A URL that points to a compressed geospatial file, possibly containing multiple Keyhole Markup Language (KML) files for the disaster. Available from 2010 to present.	yes
geoJsonUrl	GeoJSON URL	text	A URL to an API call that will return geospatial data in a GeoJSON file. Returns data for current disasters only.	yes
id	ID	uuid	Unique ID for the record that does not persist between data set refreshes	yes
hash	Hash	text	MD5 Hash of the fields and values of the record	no
lastRefresh	Last Refresh	datetimez	Date the record was last updated in the API data store	yes

Disaster Summaries

Name	Title	Type	Description	Is Searchable
disasterNumber	Disaster Number	smallint	Sequentially assigned number used to designate an event or incident declared as a disaster. For more information on the disaster process, https://www.fema.gov/disasters/how-declared	yes

totalNumberIaApproved	Total Number IA Approved	decimal	The number of disaster assistance applications that were approved for Individual Assistance (IA)	yes
totalAmountIhpApproved	Total Amount IHP Approved	decimal	The total amount approved for the Individual and Households Program (IHP) in dollars. To learn more about the programs within IHP, please read the IHP Unified Guidance: https://www.fema.gov/sites/default/files/2020-05/IHP_Unified_Guidance_FINAL_09272016_0.pdf	yes
totalAmountHaApproved	Total Amount HA Approved	decimal	The total amount approved for Housing Assistance (HA) from the Individual and Households Program (IHP) in dollars. To learn more about the programs within IHP, please read the IHP Unified Guidance: https://www.fema.gov/sites/default/files/2020-05/IHP_Unified_Guidance_FINAL_09272016_0.pdf	yes
totalAmountOnaApproved	Total Amount ONA Approved	decimal	The total amount approved for Other Needs Assistance (ONA) from the Individual and Households Program (IHP) in dollars. To learn more about the programs within IHP, please read the IHP Unified Guidance: https://www.fema.gov/sites/default/files/2020-05/IHP_Unified_Guidance_FINAL_09272016_0.pdf	yes
totalObligatedAmountPa	Total Obligated Amount PA	decimal	The Public Assistance grant funding available in dollars to the grantee (State), for sub-grantee approved Project Worksheets	yes
totalObligatedAmountCatAb	Total Obligated Amount Category A and B	decimal	The Public Assistance grant funding available in dollars to the grantee (State), for sub-grantee approved Project Worksheets in categories A and B, or Emergency Work. Category A: Debris removal; Category B: Emergency protective measures. To learn more about FEMA categories of work, visit:	yes

			https://www.fema.gov/assistance/public/program-overview	
totalObligatedAmountCatC2g	Total Obligated Amount Category C to G	decimal	The Public Assistance grant funding available in dollars to the grantee (State), for sub-grantee approved Project Worksheets in categories C through G, or Permanent Work. Category C: Roads and bridges; Category D: Water control facilities; Category E: Public buildings and contents; Category F: Public utilities; Category G: Parks, recreational, and other facilities. To learn more about FEMA categories of work, visit: https://www.fema.gov/assistance/public/program-overview	yes
paLoadDate	PA Load Date	date	The date the Public Assistance (PA) data was updated by NEMIS	yes
iaLoadDate	IA Load Date	date	The date the Individual Assistance (IA) data was updated by NEMIS	yes
totalObligatedAmountHmgp	Total Obligated Amount HMGP	decimal	Total amount obligated under the Hazard Mitigation Grant Program in dollars. The total amount obligated represents the amount obligated for Regular Project Costs, Planning Costs, Initiative Project Costs, Recipient Management Costs, Recipient Admin Costs and Subrecipient Admin Costs. To learn more about HMGP, visit: https://www.fema.gov/grants/mitigation	yes
hash	hash	text	MD5 hash of fields and values of the record	no
lastRefresh	Last Refresh	datetimez	Date the record was last updated in the API data store	yes
id	ID	uuid	Unique ID assigned to the record	yes